<u>Chapter 2: Background & Literature review</u>

<u>Optimizing playlists flow and sequencing</u>: A model to find the best path across a set of songs while maintaining a predefined "energetic arc" and ensuring high quality local transitions

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Gap Analysis & Proposed Solution: What We Do Differently

Most traditional music recommendation systems (like Collaborative Filtering or Content-Based Filtering) focus on recommending individual tracks based on semantic similarity or past user interactions, often ignoring the temporal context and the holistic listening experience. While automated DJ systems and playlist generators have been explored in the past, they typically suffer from fundamental limitations: they either focus strictly on local transition mechanics without a long-term structural goal, or they rely on generic Reinforcement Learning (RL) aiming to minimize track skips, which often fails to guarantee a cohesive musical narrative. As Spotify Researchers (2023) [1] mention in their abstract and introduction.
Our project introduces a significant innovation by formulating playlist generation as a path-finding combinatorial optimization problem on a graph, explicitly designed to maintain a predefined global "Energy Arc" (e.g., Ramp-Up or Rollercoaster) while simultaneously ensuring high-quality, seamless local transitions. By representing tracks as nodes and transition costs (calculating acoustic and harmonic disparities) as edges, we bridge the gap between global playlist dramaturgy and local acoustic compatibility, resulting in a coherent, continuous flow of energy that is explicitly tailored for party environments (at first and then the environments could expand and the system would be tailored to more genres)

1. **In-Depth Understanding of Existing Methods and Tools**
   The challenge of automated playlist generation has been approached from several distinct angles in the Music Information Retrieval (MIR) and Recommender Systems (RS) domains.
   - **Similarity-Based and Sequence Models:** Early research primarily relied on audio similarity and metadata to generate playlists, for example Bonnin et al. (2014) [2] mention the use of GPS location of users as metadata. Flexer et al. (2008) [3] demonstrated the generation of playlists by defining a start and end song, using spectral similarity to create a smooth, interpolating transition between the two anchor points. However, relying solely on similarity often results in overly homogeneous playlists that lack the dynamic variation required for an engaging listening experience (Bonnin et al. (2014) [2]), also they talked about generating a playlist while we discuss the situation where a playlist is given and we want to find fitting energetic sequences in it.

- **Graph Theory and Combinatorial Optimization:** A more sophisticated approach models track sequencing as a graph traversal problem. Bittner et al. (2016) [4] framed the optimal ordering of a playlist as finding the Shortest Hamiltonian Path in a complete weighted graph. In their model, edge weights are determined by Euclidean distances in a feature space that includes tempo (mapped logarithmically), key (mapped via the circle of fifths), and timbre. Because finding an exact Hamiltonian path is NP-complete, greedy algorithms (e.g., HAM-1, HAM-2) and Traveling Salesman Problem (TSP) approximations are widely used to minimize overall transition costs.
- **Reinforcement Learning (RL) and Markov Decision Processes (MDP):** RL has been utilized to adapt to listener preferences dynamically. Liebman et al. (2015) [5] introduced DJ-MC, an agent that models playlist recommendation as an episodic MDP, learning user preferences over both individual songs and song-to-song transitions. Recently, Spotify developed an Action-Head Deep Q-Network (AH-DQN) trained in a simulated environment to optimize user satisfaction metrics (like completion rates) by sequentially selecting tracks. [1].
- **DJ Automation and Intra-Track Analysis:** Creating a seamless mix requires analyzing the internal structure of songs to find "switch points" or cue points. Zehren et al. (2022) [6] used expert DJ rules and statistical novelty detection algorithms (like Foote's checkerboard kernel) to identify structural boundaries, such as downbeats at the start of a musical period, suitable for crossfading. Vande Veire and De Bie (2018) [7] built a complete open-source auto-DJ system for Drum and Bass that automates beat-matching, downbeat tracking, and applies specific crossfade profiles based on structural segmentation.

2. **Intelligent Comparison: Evolution and Differences Between Methods**
   The evolution of playlist generation moves from **static similarity retrieval** to **sequential optimization**, and recently toward **adaptive learning (RL).**
   - **Graph/Optimization vs. Reinforcement Learning:** Graph-based approaches (like TSP or Constraint Satisfaction Programming) are highly explainable and allow developers to enforce strict musical rules, such as preventing dissonant harmonic clashes or mandating a specific tempo trajectory. [4] However, calculating the cost matrix for large catalogs is computationally expensive ($O(N^2)$). In contrast, RL models (like DJ-MC [5]) excel at personalizing content dynamically based on user feedback (skips/listens) [1] [5]. Yet, RL models often act myopically and struggle to enforce strict long-term structural narratives, such as the classic "Hero's Journey" or "Double Peak" energy arcs used by professional DJs to prevent listener ear fatigue.

- **Inter-track vs. Intra-track Optimization:** While traditional Recommender Systems focus entirely on *inter-track mixability* (which song should play next), automated DJ research highlights the absolute necessity of *intra-track mixability* (where exactly the crossfade should occur) Zehren et al. Mentions the other sources we cite as examples for this case [6]. Studies show that misaligned beats or transitioning during vocals are primary causes of perceived poor quality in automated mixes Bittner et al. (2017) [4].

3. **Comparing Existing Solutions to Our Proposed Product** Existing commercial and academic systems largely fail to integrate the macro-level planning of a DJ set with the micro-level execution of smooth audio transitions. For example, Spotify's default shuffle or basic recommendation algorithms can create jarring, disjointed experiences because they evaluate tracks independently. While advanced RL models aim to maximize listen time [1], they do not explicitly plan an "Energy Arc" (e.g., Ramp-Up for a warmup set, or Rollercoaster for a peak-time party). **Our proposed product directly tackles this by combining Combinatorial Optimization with DJ-specific heuristics.** By calculating a unified transition cost that incorporates tempo differences, Harmonic Mixing (using the Camelot Wheel to prevent dissonant key clashes), and energy shifts, our system will search for the optimal path through the track graph. Unlike standard recommendation algorithms, our system guarantees global coherence while maintaining local transition smoothness, specifically tailoring the output for the continuous, dynamic environment of a live party.
4. **Relevant Papers, Tools, and Resources for Citation & Extraction** To build and evaluate Our system, we would utilize the following existing tools and datasets:
    - **Audio Feature Extraction Tools** (for future work and not the first part research we are doing in the scope of the course):
        - **Librosa:** The standard Python library for Music Information Retrieval (MIR). Excellent for extracting Mel-spectrograms, MFCCs, beat tracking, and onset detection.
        - **Essentia:** A highly optimized C++ library with Python bindings. It is heavily recommended over Librosa for processing large datasets quickly (up to twice as fast for tasks like FFT) and extracting advanced psychoacoustic features.
    - **Open Datasets** (what we will actually use in our project):
        - **FMA (Free Music Archive) & MTG-Jamendo:** Due to recent closures of Spotify's Audio Features API, these open datasets are critical. They provide hundreds of thousands of tracks with pre-computed acoustic features, tags, and completely open audio files suitable for rendering crossfades.
        - **Million Playlist Dataset (MPD):** Useful for analyzing human curation patterns and sequence coherence. Also mentioned to be used on Schweiger et al. (2025) [8] research on user-curated playlists.

- **UnmixDB:** Contains ground-truth annotations of DJ mixes, cue points, and beat tracking, which is essential if you plan to implement automatic crossfading and switch-point detection. Also mentioned to be used on Zehren et al. (2022) [6].

Works Cited

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